

Lecture 2

Finite element errors, fluxes, and transmission regularity

Detailed explanations accompanying the torsion manuscript

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1 Three errors that must not be confused

Aim. Turn a finite element Hessian entry into an enclosure of the continuous entry. Then investigate the transmission regularity behind its quadratic Galerkin convergence. A 90-minute lecture can cover the error identities and first-derivative regularity; the second-derivative argument below is suitable for a further reading session.

The geometry is the same regular polygon and coarse fan as in the first lecture. Write $a(v, w) = \int_P \nabla v \cdot \nabla w$ and $\|v\|_a = \|\nabla v\|_{L^2(P)}$. A displacement q has the fan extension $\theta_q = \sum_j q_j \phi_j$. These are vertex displacements, not functions that can be chosen independently on adjacent triangles. Recall that $-\Delta u = 1$, $u \in H_0^1(P)$, $\ell(v) = \int_P v$, $J = \frac{1}{2} \int_P u$, and $\mathcal{F} = J/A^2$ with $A = |P|$.

There are three distinct objects:

object	meaning	source of the difference
u	continuous weak solution	—
u_h	exact Galerkin solution in V_h	spatial discretization
$\tilde{u}$	computed coefficient vector in V_h	linear-system solve

Interval rounding is a third error, incurred while evaluating quantities in the proof. It is handled by enclosing each arithmetic operation. A small solver residual alone controls neither $u - u_h$ nor the Hessian error. Conversely, an a priori interpolation theorem does not control the rounding error in a computed residual.

Each coarse triangle is split uniformly into m^2 fine triangles. The mesh has nm^2 triangles and $1 + nm(m+1)/2$ vertices. Its boundary has nm vertices. For fixed n , $h = m^{-1}$ is proportional to the maximum fine-element diameter. All rays are unions of element edges, and the coefficients obtained by differentiating the pullback are constant on each fine triangle. A mesh crossing a ray cannot use the same broken interpolation argument without additional analysis. The space V_h consists of continuous functions that are affine on each fine triangle and zero on the polygon boundary. The exact Galerkin state is characterized by $a(u_h, v_h) = \ell(v_h)$ for every $v_h \in V_h$. It is a mathematical solution of a finite linear system, independently of how accurately a numerical solver finds its coefficients.

2 A residual is an error measured in the dual norm

For u satisfying $a(u, v) = \ell(v)$ and any conforming candidate w ,

$$a(u - w, v) = \ell(v) - a(w, v).$$

Taking the supremum over $\|v\|_a = 1$ gives an equality:

$$\|u - w\|_a = \|\ell - a(w, \cdot)\|_{(H_0^1, a)^*}.$$

The equality does not compute the dual norm. It identifies what a guaranteed majorant must bound.

For torsion, let $y \in H(\text{div}; P)$ satisfy $\text{div } y = -1$. Integration by parts, with the zero trace of v , gives

$$\ell(v) - a(w, v) = \int_P (y - \nabla w) \cdot \nabla v, \quad \|u - w\|_a \leq \|y - \nabla w\|_{L^2(P)}.$$

This is a guaranteed estimate: it is valid for every admissible y , not only the minimizer of a flux-fitting problem. It is a functional majorant in the sense discussed by Repin and by the guaranteed finite element literature [6, 5, 4].

A convenient exact construction is

$$y = -x/2 + \operatorname{curl} \psi_h, \quad \operatorname{curl} \psi_h = (\partial_y \psi_h, -\partial_x \psi_h), \quad (1)$$

where ψ_h is continuous and piecewise affine. In distributions, $\operatorname{div} \operatorname{curl} \psi_h = 0$. Across an element edge the normal component of the curl is a tangential derivative of the common trace of ψ_h , so it agrees from both sides. Thus this is a global $H(\operatorname{div})$ field with precisely the required divergence.

The numerical code chooses ψ_h by minimizing a quadratic mismatch. Its computed nodal values are simply admissible flux data. An inaccurate fit widens the bound; it does not destroy equilibration. The first part $-x/2$ is essential: a curl alone would have the wrong divergence.

Exercise 2.1. *What happens if the flux is not exactly equilibrated?*

Answer. The still-valid estimate is $\|u - w\|_a \leq \|y - \nabla w\| + C_F \|1 + \operatorname{div} y\|$, where $\|v\| \leq C_F \|v\|_a$. Exact equilibration removes the second term.

3 Differentiate the weak problem, not the approximate output

For two vertex directions, use $x \mapsto x + s\theta_q(x) + t\theta_r(x)$. The pulled-back coefficient matrix and volume density are

$$\begin{aligned} \mathbb{A}(s, t) &= \det(I + sD\theta_q + tD\theta_r)(I + sD\theta_q + tD\theta_r)^{-1}(I + sD\theta_q + tD\theta_r)^{-T}, \\ a_{s,t}(v, w) &= \int_P \nabla v^T \mathbb{A}(s, t) \nabla w, \quad \ell_{s,t}(v) = \int_P \det(I + sD\theta_q + tD\theta_r) v. \end{aligned}$$

Subscripts q, r, qr on forms mean derivatives at $s = t = 0$. In particular,

$$\begin{aligned} \mathbb{A}_q &= (\operatorname{div} \theta_q)I - D\theta_q - D\theta_q^T, \\ \ell_q(v) &= \int_P (\operatorname{div} \theta_q) v, \\ \ell_{qr}(v) &= \int_P [(\operatorname{div} \theta_q)(\operatorname{div} \theta_r) - \operatorname{tr}(D\theta_q D\theta_r)] v. \end{aligned}$$

Differentiating $a_{s,t}(u_{s,t}, v) = \ell_{s,t}(v)$ gives

$$\begin{aligned} a(u_q, v) &= \ell_q(v) - a_q(u, v), \\ a(u_{qr}, v) &= \ell_{qr}(v) - a_q(u_r, v) - a_r(u_q, v) - a_{qr}(u, v). \end{aligned} \quad (2)$$

All derivatives here are material derivatives on the fixed reference fan. They exist in H_0^1 by differentiability of the uniformly coercive solution operator; that fact alone does not give broken H^2 regularity.

Replace the test space by V_h and the state by u_h . The exact discrete first derivatives solve the resulting equations, and $z_h = u_{h,qr}$ solves the fourth discrete equation. Define the continuous lifting by

$$a(Z_{qr}^h, v) = \ell_{qr}(v) - a_q(u_{h,r}, v) - a_r(u_{h,q}, v) - a_{qr}(u_h, v). \quad (3)$$

Its Galerkin solution is z_h . The superscript h on Z_{qr}^h refers to its data; Z_{qr}^h itself is not a finite element function.

The following form bounds are finite maxima of matrix operator norms:

$$|a_q(v, w)| \leq C_q \|v\|_a \|w\|_a, \quad |a_r(v, w)| \leq C_r \|v\|_a \|w\|_a, \quad |a_{qr}(v, w)| \leq C_{qr} \|v\|_a \|w\|_a.$$

They transfer errors between the successive equations. No smallness of C_q, C_r, C_{qr} is assumed.

4 Why the second variation admits products of errors

Set $e = u - u_h$, $e_q = u_q - u_{h,q}$, and $e_r = u_r - u_{h,r}$. Galerkin orthogonality gives the exact identity

$$J - J_h = \frac{1}{2}a(e, e). \quad (4)$$

It holds on nearby polygons after pullback. Differentiating twice means differentiating the form as well as both arguments:

$$(J - J_h)_{qr} = \frac{1}{2}a_{qr}(e, e) + a_q(e_r, e) + a_r(e_q, e) + a(e_q, e_r) + a(e_{qr}, e).$$

Since $u_{h,qr} \in V_h$, orthogonality gives $a(e_{qr}, e) = a(u_{qr}, e)$. Insert the second equation in (2), tested with e , and combine the terms. The result is

$$(J - J_h)_{qr} = a(e_q, e_r) - \frac{1}{2}a_{qr}(e, e) + a(Z_{qr}^h - z_h, e). \quad (5)$$

The exact second derivative has disappeared. If

$$\|e\|_a \leq \delta_0, \quad \|e_q\|_a \leq \delta_q, \quad \|e_r\|_a \leq \delta_r, \quad \|Z_{qr}^h - z_h\|_a \leq \eta_{qr},$$

then

$$|(J - J_h)_{qr}| \leq \delta_q \delta_r + \frac{1}{2}C_{qr}\delta_0^2 + \delta_0 \eta_{qr}. \quad (6)$$

This is manuscript equation 64. It is valid on each fixed mesh without an asymptotic regularity assumption.

4.1 Could we use only material derivatives?

Yes. The weak equation already gives $a(u_q, u_r) = \ell_q(u_r) - a_q(u, u_r)$. It uses no second derivative. But replacing u by u_h changes the load, so it does not automatically remove the terms linear in the state error. Direct propagation of $O(h)$ energy bounds through the Hessian generally leaves an $O(h)$ bound. The Bogosel–Bucur analysis already exploits a quadratic interpolation product for material derivatives; the leading first-order contribution comes from other data errors, not a failure of Galerkin orthogonality [2, Section 3.1].

In (6), all four errors of order h instead produce an order- h^2 bound. This improvement is due to the cancellation and lifting. Equilibrated fluxes are a way to enclose the energy errors; they are not, by themselves, a theorem of quadratic convergence.

5 The computed linear systems also need error bounds

Let $Kx_* = f$ be the exact interior-node system, with K symmetric positive definite. A computed vector x has residual $f - Kx$. If $\lambda_{\min}(K) \geq \underline{\alpha} > 0$, then

$$\|x - x_*\|_K^2 = (f - Kx)^T K^{-1} (f - Kx) \leq \frac{\|f - Kx\|_2^2}{\underline{\alpha}}.$$

The Euclidean norm of the residual is not the energy norm of the error; the factor $1/\sqrt{\underline{\alpha}}$ supplies that conversion.

For the circumradius-one polygon, domain monotonicity and the square give $\lambda_1(P) \geq \pi^2/2$. On one triangle,

$$M_T = \frac{|T|}{12} \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix} \succeq \frac{|T|}{12} I.$$

After assembly and restriction to interior nodes,

$$\underline{\alpha} = \frac{\pi^2}{24} \min_{i \text{ interior}} \sum_{T \ni i} |T| \quad \text{is a valid lower bound for } \lambda_{\min}(K). \quad (7)$$

This is explicit and possibly conservative. It contains no numerically estimated inverse and no uncertified eigenvalue computation.

Define the four residual functionals of the computed candidates:

$$\begin{aligned} \rho_0(v) &= \ell(v) - a(\tilde{u}, v), \\ \rho_q(v) &= \ell_q(v) - a_q(\tilde{u}, v) - a(\tilde{u}_q, v), \\ \rho_r(v) &= \ell_r(v) - a_r(\tilde{u}, v) - a(\tilde{u}_r, v), \\ \rho_z(v) &= \ell_{qr}(v) - a_q(\tilde{u}_r, v) - a_r(\tilde{u}_q, v) - a_{qr}(\tilde{u}, v) - a(\tilde{z}, v). \end{aligned}$$

Their vectors are obtained by evaluating them on each interior basis function φ_i : $(r_\rho)_i = \rho(\varphi_i)$. For example, $r_{\rho_0} = f - K\tilde{x}$.

Because later loads contain earlier unknowns, the transfer is

$$\begin{aligned} \epsilon_0 &= \|r_{\rho_0}\|_2 / \sqrt{\underline{\alpha}}, \\ \epsilon_q &= \|r_{\rho_q}\|_2 / \sqrt{\underline{\alpha}} + C_q \epsilon_0, \\ \epsilon_r &= \|r_{\rho_r}\|_2 / \sqrt{\underline{\alpha}} + C_r \epsilon_0. \end{aligned}$$

These bound the differences between exact discrete solutions and their candidates. Boundary coefficients are set to exactly zero before any residual is assembled. A large penalty imposed by a floating-point solver is not a substitute for this Dirichlet restriction.

5.1 Correct the centre before increasing the radius

The energy centre $\tilde{J}_{\text{corr}} = \ell(\tilde{u}) - a(\tilde{u}, \tilde{u})/2$ satisfies $J_h - \tilde{J}_{\text{corr}} = \|u_h - \tilde{u}\|_a^2/2$. For the mixed derivative use

$$\begin{aligned} \tilde{C}_{\text{corr}} &= \ell_{qr}(\tilde{u}) + \ell_q(\tilde{u}_r) - \frac{1}{2}a_{qr}(\tilde{u}, \tilde{u}) - a_q(\tilde{u}, \tilde{u}_r) \\ &\quad + \rho_r(\tilde{u}_q) + \rho_0(\tilde{z}). \end{aligned}$$

Expanding about the exact discrete fields and substituting the residual identities gives

$$\begin{aligned} J_{h,qr} - \tilde{C}_{\text{corr}} &= \rho_z(u_h - \tilde{u}) - \frac{1}{2}a_{qr}(u_h - \tilde{u}, u_h - \tilde{u}) \\ &\quad + a(u_{h,q} - \tilde{u}_q, u_{h,r} - \tilde{u}_r). \end{aligned}$$

Every first-order material-solve term has cancelled. Consequently

$$|J_{h,qr} - \tilde{C}_{\text{corr}}| \leq E_{\text{alg}} := \frac{\|r_{\rho_z}\|_2}{\sqrt{\underline{\alpha}}} \epsilon_0 + \frac{1}{2} C_{qr} \epsilon_0^2 + \epsilon_q \epsilon_r.$$

The lifting solve error need not be given another separate name: its residual appears directly in this expression.

6 Four fluxes and the final continuous enclosure

Choose four continuous affine potentials. The state flux is (1); the first-derivative flux is $-\theta_q + \text{curl } \psi_{q,h}$, with divergence $-\text{div } \theta_q$. For the second lifting use

$$-(\text{div } \theta_q) \theta_r + D \theta_q \theta_r + \text{curl } \psi_{z,h}.$$

Indeed,

$$\operatorname{div}((\operatorname{div} \theta_q) \theta_r - D\theta_q \theta_r) = (\operatorname{div} \theta_q)(\operatorname{div} \theta_r) - \operatorname{tr}(D\theta_q D\theta_r).$$

Its normal trace is continuous across fan edges: the gradient jump of a continuous affine field is a vector tensor the edge normal, and the two normal jump contributions cancel. This is a distributional identity, not only a trianglewise identity.

The four mismatch norms are

$$\begin{aligned}\Phi_0 &= \| -x/2 + \operatorname{curl} \psi_{0,h} - \nabla \tilde{u} \|, \\ \Phi_q &= \| -\theta_q + \operatorname{curl} \psi_{q,h} - \nabla \tilde{u}_q - \mathbb{A}_q \nabla \tilde{u} \|, \\ \Phi_r &= \| -\theta_r + \operatorname{curl} \psi_{r,h} - \nabla \tilde{u}_r - \mathbb{A}_r \nabla \tilde{u} \|, \\ \Phi_z &= \| -(\operatorname{div} \theta_q) \theta_r + D\theta_q \theta_r + \operatorname{curl} \psi_{z,h} - \nabla \tilde{z} \\ &\quad - \mathbb{A}_q \nabla \tilde{u}_r - \mathbb{A}_r \nabla \tilde{u}_q - \mathbb{A}_{qr} \nabla \tilde{u} \|.\end{aligned}$$

All unmarked norms in these formulas are $L^2(P)$ norms of vector fields. Residual bounds, load-error transfer, and best approximation give

$$\begin{aligned}\hat{\delta}_0 &= \Phi_0, \\ \hat{\delta}_q &= \Phi_q + C_q \Phi_0 + \epsilon_q, & \hat{\delta}_r &= \Phi_r + C_r \Phi_0 + \epsilon_r, \\ \hat{\eta}_{qr} &= \Phi_z + C_q \epsilon_r + C_r \epsilon_q + C_{qr} \epsilon_0.\end{aligned}$$

For example, Φ_0 bounds the error to $\tilde{u}$; best approximation then also bounds the error to u_h . In the lifting bound, $\tilde{z} \in V_h$ can replace z_h before best approximation is used. This explains why an extra lifting algebraic error is not added there.

For a nonzero Fourier pairing, $A_q = A_r = 0$. Define

$$B_J = \hat{\delta}_q \hat{\delta}_r + \frac{1}{2} C_{qr} \hat{\delta}_0^2 + \hat{\delta}_0 \hat{\eta}_{qr}.$$

The entry centre and radius are

$$\begin{aligned}\tilde{F}_{qr} &= \frac{\tilde{C}_{\text{corr}}}{A^2} - \frac{2\tilde{J}_{\text{corr}} A_{qr}}{A^3}, \\ R_F &= \frac{B_J + E_{\text{alg}}}{A^2} + \frac{|A_{qr}|}{A^3} (\hat{\delta}_0^2 + \epsilon_0^2), \\ |D^2 \mathcal{F}[q, r] - \tilde{F}_{qr}| &\leq R_F.\end{aligned}\tag{8}$$

Every term has a job: B_J is the continuous discretization error, E_{alg} is the error in the discrete centre, and the last term accounts for the area-normalization coefficient multiplying J . If the evaluated centre is an interval, its arithmetic radius is retained.

7 Regularity proof: integrate the vertex-zero load by parts

This section develops the proposed Bogosel–Bucur route. We will prove broken regularity of first and second material derivatives *at the regular polygon*. The positive shift is justified by a compatible trace lifting and reflection, followed by the Dirichlet polygonal shift theorem. The manuscript now proves this fixed-polygon result in Proposition 6.2 and its quadratic Hessian rate in Theorem 6.3. Uniformity over perturbed polygons remains a separate question.

For $c = 1, 2$, put the weak equation for U_0^c in the form

$$\begin{aligned}a(U_0^c, v) &= \int_P G_{0,c}(u) \cdot \nabla v + \int_P (\partial_c \phi_0) v, \\ G_{0,c}(u) &= -(\partial_c \phi_0) \nabla u + (\partial_c u) \nabla \phi_0 + (\nabla \phi_0 \cdot \nabla u) e_c.\end{aligned}\tag{9}$$

On a sector the hat gradient is constant. Since $\Delta u = -1$,

$$\operatorname{div} G_{0,c}(u) = \partial_c \phi_0 + 2\nabla \phi_0 \cdot \nabla \partial_c u.$$

The volume load after integration by parts is therefore

$$-2\nabla \phi_0 \cdot \nabla \partial_c u \in L^2(P). \quad (10)$$

It is supported on $T_{-1} \cup T_0$. The term $\partial_c \phi_0$ has cancelled; keeping it would give an incorrect strong formulation.

Let $S_j = \{ra_j : 0 < r < 1\}$ and let $\nu_j = (-\sin(j\vartheta), \cos(j\vartheta))$ point from T_{j-1} into T_j . The trace convention is $[w]_j = w|_{T_j} - w|_{T_{j-1}}$. The contribution of the two boundary integrals is $-[G_{0,c}(u)]_j \cdot \nu_j$. Continuity of ϕ_0 implies $[\nabla \phi_0]_j$ is normal to S_j . Because $u \in H^2(P)$, its gradient has a common trace there. Substituting into (9) gives the especially simple ray density

$$-[G_{0,c}(u)]_j \cdot \nu_j = -([\nabla \phi_0]_j \cdot \nu_j) \partial_c u|_{S_j}. \quad (11)$$

Only S_{-1}, S_0, S_1 occur. Their coefficients are

$$[\nabla \phi_0]_0 \cdot \nu_0 = -2 \cot \vartheta, \quad [\nabla \phi_0]_{\pm 1} \cdot \nu_{\pm 1} = \frac{1}{\sin \vartheta}.$$

Thus the complete distributional equation is

$$-\Delta U_0^c = -2\nabla \phi_0 \cdot \nabla \partial_c u + 2 \cot \vartheta (\partial_c u)|_{S_0} \delta_{S_0} - \frac{(\partial_c u)|_{S_1} \delta_{S_1} + (\partial_c u)|_{S_{-1}} \delta_{S_{-1}}}{\sin \vartheta}. \quad (12)$$

Here $g\delta_S$ means the functional $v \mapsto \int_S gv \, ds$. No point masses occur: we integrated bounded piecewise coefficients times H^1 gradient fields once, and the singular terms are edge measures.

7.1 Why the ray densities have the right endpoint behavior

For fixed n , convex-polygon Dirichlet regularity gives

$$u \in H^{2+\sigma}(P), \quad 0 < \sigma < \min \left\{ \frac{1}{2}, \frac{\pi}{\pi - 2\pi/n} - 1 \right\},$$

after decreasing σ if necessary. Possible resonant logarithmic corner terms are harmless below this strict exponent threshold. In particular ∇u is continuous up to each corner. Its tangential derivatives on the two adjacent Dirichlet edges are zero; the tangents are independent, so $\nabla u(a_j) = 0$. At the centre, rotational symmetry implies $\nabla u(0) = 0$.

On S_j , reflection gives zero normal derivative, hence

$$\nabla u(ra_j) = a_j \partial_r u(r, 0).$$

Thus the densities in (12) are precisely rotated multiples of the same radial derivative. The trace theorem gives $H^{1/2+\sigma}$ regularity on each ray, and the values at $r = 0, 1$ vanish. Their extensions by zero across the centre belong to the same fractional space for sufficiently small $\sigma > 0$. Endpoint values here are justified by the extra positive regularity; bare $H^{1/2}$ membership would not by itself justify a pointwise endpoint argument.

7.2 The mixed-boundary ingredient, with its hypotheses visible

Lemma 7.1 (A ray load with vanishing endpoints). *Fix a regular polygon. Let $g \in H^{1/2+\sigma}(0, 1)$ with $g(0) = g(1) = 0$ and $\sigma > 0$ sufficiently small. The solution $W \in H_0^1(P)$ of $a(W, v) = \int_{S_0} gv$ belongs to $H^{2+\sigma'}$ on each side of the ray and on every fan sector, for some $\sigma' > 0$ that may be smaller than σ . The same assertion holds for any rotated ray.*

Proof. The line load is continuous on $H_0^1(P)$ by the trace theorem, so Lax–Milgram gives W . Its data and domain are invariant under reflection across the line through S_0 , hence W is even. Restrict to the upper half-polygon. The boundary conditions are Dirichlet zero on the polygon boundary and outward Neumann data $g/2$ on the positive part of the cut, zero on its negative part. The factor $1/2$ follows by doubling the weak integral over the half-polygon.

To justify a positive shift, lift the nonzero Neumann data before reflecting. Choose $0 < \sigma' < \min\{\sigma, 1/2, 2/(n-2)\}$. Prescribe zero value on the entire half-polygon boundary, outward normal derivative $g/2$ on the loaded cut, and zero normal derivative on all remaining sides. Theorem 1 of Bernardi–Dauge–Maday [1], with $p = 2$, $s = 2 + \sigma'$, and two traces, gives a lifting $v_0 \in H^{2+\sigma'}$. The corner compatibility conditions are precisely agreement of values and first gradients here. They hold because the prescribed value and both endpoint normal derivatives are zero. Zero extension through the centre preserves the required $H^{1/2+\sigma'}$ regularity of the cut data.

The difference $W - v_0$ has homogeneous Neumann data on the cut. Its even reflection solves a Dirichlet problem on the full polygon, with right-hand side the even extension of Δv_0 . This belongs to $H^{\sigma'}$ since $\sigma' < 1/2$. The Dirichlet shift theorem [3, Chapter 4] gives $H^{2+\sigma'}$ regularity because $1 + \sigma' < n/(n-2)$, the first Dirichlet corner exponent. Adding back v_0 proves the assertion on each half-polygon. This supplies the positive shift needed below; an H^2 estimate alone would not suffice.

For odd n , the negative cut ends at an edge midpoint with zero Neumann data nearby; reflection gives a straight Dirichlet boundary there. Across the interior of that negative cut, W is harmonic and its first derivatives match. Its piecewise $H^{\sigma'}$ second derivatives glue for $\sigma' < 1/2$, so the same regularity holds on each original fan sector even when the cut crosses a sector. The ray decomposition follows [2, Section 3]; the trace lifting above makes the endpoint compatibility explicit. $\square$

Proposition 7.2 (First material derivatives at the regular polygon). *For every fixed $n \geq 3$, the first vertex material derivatives are in $H^{2+\sigma}$ on each coarse fan triangle for some $\sigma > 0$ depending on n . Their gradients have a common value at the centre and vanish at the polygon vertices.*

Proof. Split (12) into its volume solution and three ray solutions. The volume load is piecewise H^σ , hence globally H^σ after choosing $\sigma < 1/2$; multiplication by a sector indicator preserves H^σ in this range. The convex Dirichlet polygonal shift theorem gives $H^{2+\sigma}$ for its solution, with a possibly smaller positive exponent. Lemma 7.1 handles the three ray solutions. Rotation and linear combination handle all vertices and all directions q .

The one-sided gradients are continuous to sector endpoints by Sobolev embedding. Across a ray, the tangential gradient agrees, because the function has one common trace. The normal-gradient jump equals the line density, up to the chosen orientation. That density vanishes at both endpoints, so the gradients agree at the centre and at each outer endpoint. At a polygon vertex the common gradient is orthogonal to both adjacent boundary tangents, hence is zero. At the centre the common gradient need not be zero; symmetry of the state does not imply the same symmetry for a single-vertex material derivative. $\square$

8 The second derivatives: a necessary centre cancellation

The first-derivative proof is not automatically a proof for u_{qr} . Nevertheless it supplies the extra regularity needed to continue. On each sector write its weak equation as

$$a(u_{qr}, v) = \int_P d_{qr} v + \int_P G_{qr} \cdot \nabla v,$$

where, for this regularity calculation only,

$$\begin{aligned} d_{qr} &= (\operatorname{div} \theta_q)(\operatorname{div} \theta_r) - \operatorname{tr}(D\theta_q D\theta_r), \\ G_{qr} &= -\mathbb{A}_q \nabla u_r - \mathbb{A}_r \nabla u_q - \mathbb{A}_{qr} \nabla u. \end{aligned}$$

These two names display the volume and flux parts of one distribution; they do not introduce any new PDE solves. Sectorwise integration by parts produces $d_{qr} - \operatorname{div} G_{qr} \in L^2$ and ray densities $-[G_{qr}]_j \cdot \nu_j \in H^{1/2+\sigma}(S_j)$. All outer endpoint values vanish by Proposition 7.2. The centre values may not vanish because $\nabla u_q(0)$ and $\nabla u_r(0)$ may be nonzero.

This matters. A constant load on a ray ending at an interior point can produce an $r \log r$ corner term; its second derivatives are not square integrable near that endpoint. Splitting a compatible collection of rays into unrelated single-ray problems can therefore destroy the regularity one is trying to prove.

8.1 An explicit local lifting removes the centre values

Choose a smooth cutoff χ supported in an interior disk around the centre, equal to one in a smaller disk. Subtract from u_{qr} the known piecewise smooth function

$$\chi(x)(\theta_q(x) \cdot \nabla u_r(0) + \theta_r(x) \cdot \nabla u_q(0)). \quad (13)$$

It belongs to $H_0^1(P)$ and is H^2 on each fan sector. Its purpose is local regularity analysis; it is not the numerical second lifting Z_{qr}^h . The centre gradients are well defined by the first-derivative proposition.

For any constant vector b , the following identity holds in distributions:

$$\operatorname{div}(-\mathbb{A}_q b) = \Delta(\theta_q \cdot b). \quad (14)$$

Indeed, $-\mathbb{A}_q b = D\theta_q b + D\theta_q^T b - (\operatorname{div} \theta_q)b$. The divergences of the first and last terms cancel by commutation of weak derivatives; the middle term is $\nabla(\theta_q \cdot b)$. Near the centre, where $\chi = 1$, identity (14) shows that (13) has exactly the ray-jump contributions of $-\mathbb{A}_q \nabla u_r(0) - \mathbb{A}_r \nabla u_q(0)$. Also $\nabla u(0) = 0$, so the $\mathbb{A}_{qr}$ term contributes no centre value. Subtracting the lifting therefore makes every remaining ray density vanish at the centre.

Differentiating the cutoff introduces only piecewise smooth volume terms and ray terms supported away from the centre and the outer vertices. The resulting volume source is still L^2 ; each ray density is still $H^{1/2+\sigma}$ and now vanishes at *both* endpoints. The regular volume problem and the isolated ray problems can now be treated separately.

Proposition 8.1 (Second material derivatives at the regular polygon). *The exact second material derivative u_{qr} belongs to $H^2(T_j)$ for every fan triangle, every fixed regular polygon, and every pair of vertex directions q, r .*

Proof. After subtracting (13), solve the L^2 volume problem with zero Dirichlet data. Its solution is globally H^2 by convexity. For each remaining ray load apply Lemma 7.1. The sum of these solutions and the subtracted lifting satisfies the weak equation for u_{qr} , and uniqueness identifies it with u_{qr} . Every summand is H^2 on each fan triangle. There are finitely many rays and finitely many vertex coordinates, so the argument applies to all mixed directions. $\square$

8.2 Scope of the regularity and rate theorems

The argument proves the qualitative broken regularity of the state, first derivatives, and second derivatives *at* P_n , for fixed n . It also gives uniformity over bounded vertex directions there, by finite dimensionality and bilinearity. Constants may depend on n ; the corner exponent tends to one as n grows.

The manuscript's Theorem 6.3 now states the Hessian rate at this fixed regular polygon. Uniform broken regularity in a neighborhood of perturbed polygons is a stronger statement: those polygons no longer have the reflection symmetries used here. Such an extension would need stability of the transmission estimates under geometric perturbations, or a parameter-dependent corner expansion with uniform control.

For the Hessian rate *at the regular polygon*, the regularity just proved is sufficient. Broken interpolation and the differentiated Galerkin equations give $\delta_0, \delta_q, \delta_r = O(h)$. Moreover, subtracting the exact second-derivative equation from (3) gives

$$\|Z_{qr}^h - u_{qr}\|_a \leq C_q \delta_r + C_r \delta_q + C_{qr} \delta_0 = O(h).$$

Approximate the fixed function u_{qr} by its broken interpolant and use best approximation for Z_{qr}^h to obtain $\eta_{qr} = O(h)$. Equation (6) then gives $D^2 J(P_n) - D^2 J_h(P_n) = O(h^2)$. The energy identity and its first derivative also bound $J - J_h$ and $D(J - J_h)$ by $O(h^2)$, so differentiating A^{-2} gives the same rate for the normalized Hessian. This is a statement about exact Galerkin quantities, not yet about the computed radius in (8).

Exercises and proof checkpoints

Exercise 8.2. For $c = 2$, simplify the three ray terms in (12) using $\nabla u(ra_j) = a_j \partial_r u(r, 0)$.

Answer. The S_0 term is zero. The terms on S_1, S_{-1} are respectively $-\partial_r u \delta_{S_1}$ and $+\partial_r u \delta_{S_{-1}}$, with the traces understood on their rays. For $c = 1$ the weights are $2 \cot \vartheta, -\cot \vartheta, -\cot \vartheta$.

Exercise 8.3. Why is pointwise zero at an endpoint not an argument for a general $H^{1/2}$ function?

Answer. Endpoint evaluation is not a continuous trace at the borderline exponent. The positive extra regularity above, or an explicit weighted endpoint condition, must be used.

Exercise 8.4. Which extra facts are needed to conclude that the computed R_F is $O(h^2)$?

Answer. The four chosen flux mismatches and the transferred algebraic bounds must have compatible orders, so that B_J, E_{alg} , and the normalization remainder are $O(h^2)$. A fixed solver tolerance or fixed precision can eventually prevent that asymptotic behavior even when the exact Galerkin Hessian converges quadratically.

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